

Hydraulic conductivity of the root-soil interface of lupin in sandy soil after drying and rewetting

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Abstract

Aims The putative role of the rhizosphere in controlling root water uptake is receiving increasing attention. Recent experiments showed that the rhizosphere turned temporarily hydrophobic after drying and subsequent rewetting. Our objective was to investigate whether the rhizosphere hydrophobicity influences the hydraulic conductivity of the rhizosphere-root continuum.

Methods Lupins were grown in aluminium containers filled with a sandy soil. When the plants were 30 days-old, the soil was let dry to a water content of 2–4 % and then it was irrigated. The soil water content during irrigation was imaged using a time-series neutron radiography. By image processing, we quantified the increase in the volume of water in the roots upon irrigation. Using this information and additional measurements of the root pressure after irrigation, we calculated the water flow into the roots and the total hydraulic conductance of the rhizosphere-root continuum, K_{tot} .

Results The radiographs showed that: 1) the rhizosphere stayed temporarily dry after irrigation; 2) as the rhizosphere slowly rewetted, the roots rehydrated of 63 %. During 2 to 3 h subsequent to irrigation, K_{tot} increased from $1.36 \pm 1.09 \times 10^{-11} \text{ m}^2 \text{ s}^{-1} \text{ MPa}^{-1}$ to $5.02 \pm 2.13 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ MPa}^{-1}$, approaching the conductance of lupin roots in wet soils measured in previous experiments. Based on our calculations, these values of K_{tot} correspond to a rhizosphere conductivity of $3.87 \pm 1.91 \times 10^{-14} \text{ m s}^{-1}$ (shortly after irrigation) and $1.09 \pm 1.64 \times 10^{-12} \text{ m s}^{-1}$ (2–3 h after irrigation).

Conclusions We conclude that during a drying/wetting cycle, the conductivity of the root-soil interface is a temporary limit to root water uptake. We postulate that the temporary reduced hydraulic conductivity is primarily caused by the rhizosphere water repellency.

Keywords Hydraulic conductivity · Hydrophobicity · Neutron radiography · Rhizosphere · Root pressure probe · Root-soil interface · Root water uptake

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Introduction

Rate and location of root water uptake are influenced by the hydraulic conductivity of the roots and as the soil dries by the hydraulic conductivity of the soil in the vicinity of the roots, the so-called *rhizosphere* (Passioura 1988; Hallett et al. 2003). Both conductivities vary over time and in space and are influenced by complex processes and root-soil interactions as briefly introduced in the next paragraphs.

The root hydraulic conductivity changes during the developmental stages of plants. It also changes in response to external conditions such as soil drying and water stress (Nobel and Cui 1992a, 1992b; Enstone et al. 2002). Some of the changes in root hydraulic conductivity are irreversible, like the formation and development of casparian bands and deposition of suberin lamellae at the endodermis. In contrast, some of the changes in root hydraulic conductivity are reversible, like the expression and regulation of aquaporins (Bramley et al. 2009; Knipfer et al. 2011; McLean et al. 2011). These adaptations are thought to be a strategy by plants to improve water and nutrient uptake.

The soil hydraulic conductivity decreases several orders of magnitude as the soil dries. Additionally, as the soil dries, the hydraulic conductivity of the root-soil interface is affected by the root shrinkage and formation of air-filled gaps at the root-soil interface (Nobel and Cui 1992b; North and Nobel 1997; Carminati et al. 2009) and soil compaction due to root growth (Aravena et al. 2011). It has been shown that due to the radial geometry of the water flow to the roots, root water uptake is particularly influenced by the hydraulic conductivity of the soil near the roots (Van Lier et al. 2006; Carminati et al. 2011).

The hydraulic properties of the rhizosphere are subjected to several dynamic modifications. Roots and microorganisms living in symbiosis with roots continuously modify the hydraulic properties of the rhizosphere (Gregory and Hinsinger 1999; Hallett et al. 2003). Indeed, an increasing number of studies on soil structure, soil aggregation and water distribution around the roots showed that the physical properties of the rhizosphere differ in several ways from those of the bulk soil (Watt et al. 1994; Young 1995; Carminati et al. 2010; Moradi et al. 2011).

In particular, the presence of mucilage, a polymeric gel that is exuded by most plant roots, is believed to be one of the main factors influencing the rhizosphere hydraulic properties. A variety of functions have been attributed to mucilage: maintenance of a good contact between roots and soil particles, reduction of friction for root penetration, facilitation of root water uptake by increasing the rhizosphere water content, and avoidance of root tissue dehydration while the root tip is emerging (McCully 1995; Hallett et al. 2003; Carminati et al. 2011; Ahmed et al. 2014).

Mucilage is a hydrogel secreted by the root tip in soil. Mucilage can absorb large volumes of water (McCully

and Boyer 1997), but it also contains a small fraction of amphiphilic components such as lipids (Read et al. 2003). Under dry conditions, the hydrophilic ligands of these components may bind to each other or to the soil and expose the hydrophobic parts towards the air-filled phase of the pore space, causing water repellency in the rhizosphere. Rhizosphere water repellency was observed in barley (Hallett et al. 2003) and lupins (Moradi et al. 2011). Although it has been shown that the rhizosphere turned hydrophobic after drying and rewetting (Carminati et al. 2010; Moradi et al. 2012; Carminati and Vetterlein 2013), to date, it has been technically difficult to quantify the effect of this hydrophobicity on root water uptake. This lack of experimental data is mainly due to the difficulty to measure the hydraulic properties of the root-soil interface *in-situ* and in controlled conditions. However, such information is urgently required for a better understanding and modeling of water flow from soil to the roots.

We previously showed that the water flow into roots was reduced as the rhizosphere became hydrophobic (Zarebanadkouki and Carminati 2014). We postulated that the reduced flow of water was caused by the low hydraulic conductivity of the rhizosphere during the rewetting phase after severe drying.

To test this hypothesis, we introduce here a technique to estimate the effective hydraulic conductivity of the root-rhizosphere continuum after irrigation subsequent to a drying cycle. We monitored the distribution of water in soil and roots after irrigation by means of neutron radiography. Based on the root swelling, we calculated the water flow to the root. The experiments were conducted at nighttime, when transpiration was negligible. We also monitored the root pressure during and after irrigation using a root pressure probe that was connected to the plants. From these data, we calculated the effective hydraulic conductivity of the root-rhizosphere continuum. The results of this study will help to better understand and model the effects of the rhizosphere on soil and plant water relations.

Materials and methods

Preparation of soil and plants

Lupin seeds (*Lupin Albus*) were grown in aluminum containers (15 cm wide, 30 cm high and 1 cm thick) filled with a sandy soil. The soil was collected from the

artificial catchment of Chicken Creek near Cottbus, Germany. The soil (sieved to <2 mm) consisted of approximately 92 % sand, 5 % silt and 3 % clay. The hydraulic properties of the soil are reported in Table 1 and in the supplementary information. The aluminum containers were placed horizontally on the table with one large side left open to enable the filling of the containers. The air-dry sandy soil was poured into the containers through a sieve (mesh of 2 mm) to favor a homogenous sand deposition and limit soil layering. Then, the open sides of the containers were closed and the containers were turned vertically and gently shaken to achieve a stable packing. The average bulk density of the packed soil was 1.4 g cm^{-3} . The top of the samples was covered with a 1 cm-thick layer of quartz gravel (grain diameter of 2 mm) to minimize evaporation. The containers had holes at the bottom that allowed irrigation and drainage from the bottom.

Lupin seeds were germinated on moist filter paper for 24 h. One seedling was then planted into the soil at 1 cm depth of each container. The plants were grown with a photoperiod of 14 h, light intensity of $300 \mu\text{mol m}^{-2} \text{ s}^{-1}$, day/night temperature of $25^\circ\text{C}/19^\circ\text{C}$, and relative humidity of 60 %. During the first 4 days after planting, the samples were irrigated every day from the top. Afterwards, we irrigated the plants every fourth day by slowly immersing the samples in a water reservoir until the water table reached at a height of 15 cm above the bottom of the sample. The samples were left in the water reservoir for 30 min, then they were removed from the water reservoir and they were let drain for additional 60 min. After this procedure the average soil water content in the samples was approximately $0.30 \text{ m}^3 \text{ m}^{-3}$.

Transpiration was monitored gravimetrically by weighing the samples every 8 h. The average daily transpiration rate was 1.04 ± 0.13 ($n=6$) g h^{-1} per plant (3 weeks old plants). Twenty five days after planting, we

stopped the irrigation and let the plants dry the soil. The transpiration rate stayed rather constant for 4 days after stopping the irrigation, and then it reduced to 0.4 ± 0.21 ($n=6$) g h^{-1} (measured during daytime) when we started the neutron radiography experiments. At this point the plants were 30 days old and the average soil water content was $0.02\text{--}0.04 \text{ m}^3 \text{ m}^{-3}$.

Rewetting experiment

The neutron radiography measurements began 6 days after irrigation was stopped. At this point, the plants were 30 days old. The plants were then irrigated and a time-series neutron radiography was used to monitor water redistribution in the soil and the roots. A peristaltic pump was used to irrigate three samples from the top with a constant flow rate of 1 ml min^{-1} . This flow rate corresponded to a flow velocity smaller than the saturated hydraulic conductivity of the soil. We refer to this experiment as constant irrigation rate experiment. To favor a uniform wetting front through the sample, the irrigation was performed at six locations with even intervals. Irrigation continued for 3 h in order to maintain the water content between 0.19 and $0.24 \text{ m}^3 \text{ m}^{-3}$ during the measurements. Note that the soil matric potential in this range of water contents is between 3 and 4.5 kPa, as estimated based on the water retention curve (see [supplementary information](#)).

In a second experimental set-up, two samples were irrigated with a flow rate of 10 ml min^{-1} for a period of 6 min and then the irrigation was stopped. We refer to this experiment as pulse irrigation experiment.

In both cases, the samples were irrigated when the soil was at a water content of $0.02\text{--}0.04 \text{ m}^3 \text{ m}^{-3}$. Based on the measured water retention curve ([supplementary information](#)), these water contents correspond to soil matric water potentials of -50 to -100 kPa .

The spatiotemporal distribution of water during irrigation was monitored using neutron radiography. The experiments were performed at nighttime when transpiration was negligible.

Neutron radiography

Neutron radiography is an excellent technique to quantitatively image the distribution of roots and water in thin soil samples (Moradi et al. 2009). Neutron radiography was carried out at the cold neutron imaging beamline ICON at the Paul Scherrer Institute (PSI),

Table 1 van Genuchten parameters of the water retention curve. The parameters were obtained by fitting the data of soil water content at eight different matric potentials; the measurements were done on disturbed samples using a pressure plate apparatus. The samples were cylinders of 6 cm in height and 6 cm in diameter. θ_r is the residual water content, θ_s is the saturated water content, n and α are fitting parameters and K_s is the saturated soil hydraulic conductivity

$\theta_r [\text{cm}^3 \text{ cm}^{-3}]$	$\theta_s [\text{cm}^3 \text{ cm}^{-3}]$	$n [-]$	$\alpha [\text{cm}^{-1}]$	$K_s [\text{cm s}^{-1}]$
0.015	0.41	1.91	0.047	0.003

Switzerland. We used a CCD camera detector with an array of 1260×1260 pixels, resulting in a field of view of 15.75×15.75 cm, and pixel size of 0.013 mm. Two radiographs with marginal overlaps were needed to scan the entire sample.

Image processing

The relation between water content and neutron attenuation is given by the Beer-Lambert law:

$$-\log\left(\frac{I(x,y,t)-dc}{I_{dry}-dc}\right) = \mu_{H_2O}d_{H_2O}(x,y,t) \quad (1)$$

where x and y are the spatial coordinates of the field of view, t is time, $I(x,y,t)$ is the transmitted beam intensity, I_{dry} is the transmitted beam intensity through the container filled with dry soil, and dc is the dark current (an image recorded by the camera when there beam was off). The assumption is that the soil density did not change during irrigation. The variable $d_{H_2O}(x,y,t)$ is the thickness of water in the beam direction and $\mu_{H_2O} = 3.65 \text{ cm}^{-1}$ is the neutron attenuation coefficient of water. The value of μ_{H_2O} was calculated using a control sample with a known thickness, which was filled with normal water.

The sharp contrast between the roots and the surrounding soil allowed us to segment the roots from the soil. Roots were segmented using the algorithm Roottracker 2D developed by Anders Kaestner and described in Menon et al. (2007). We selected root segments that did not overlap with each other. All the selected root segments had a distance of 5 to 10 cm from the root tip. Root length and root radius were calculated using the Euclidean distance map and the skeleton of the segmented roots in matlab. After root segmentation, we calculated the water content in soil and roots as follows: in the pixels where there was no root, the average soil water content $\bar{\theta}(x,y,t)$ [cm cm^{-1}] was calculated as:

$$\bar{\theta}(x,y,t) = \frac{d_{H_2O}(x,y,t)}{d_{tot}} \quad (2)$$

where $d_{tot} = 1$ cm is the thickness of the inner space of the container in the beam direction.

The water content in the rhizosphere was calculated as a function of distance from the root surface. Since the radiographs are 2D images resulting from the neutron transmission across the thickness of samples, the pixels next to the root in the radiographs give the average water

content in the rhizosphere and in the portion of the soil in front of and behind of the rhizosphere. To obtain the actual water content in the rhizosphere, we assumed that the water distribution around the roots was radially symmetric. In this case, the relation between the actual water content as a function of distance from the root center, $\theta(r)$, and the average water content as a function of distance from the root surface in the images, $\bar{\theta}(x)$, is given by:

$$\bar{\theta}(x) = \frac{2}{d_{tot}} \int_x^{\sqrt{x^2 + \left(\frac{d_{tot}}{2}\right)^2}} \theta(r) \sqrt{\frac{r^2}{r^2 - x^2}} dr \quad (3)$$

where d_{tot} is the inner thickness of the sample, r is the radial distance from the root center, and x is the apparent distance from the root center in the radiograph (Fig. 1). Equation (3) is derived by changing the coordinate into radial coordinate. The average signal measured at the position x on the detector is the average of the actual water content between the distances of x and $\sqrt{x^2 + \left(\frac{d_{tot}}{2}\right)^2}$ from the root surface, which are the limits of integration in Eq. (3). The term $\sqrt{\frac{r^2}{r^2 - x^2}}$ comes from the transformation into the radial coordinates.

In the pixels containing a root, the neutron attenuation is the average of the attenuation coefficients of the root and of the soil in front and behind it – i.e. the root occupies only a few millimeters of the sample thickness in the beam direction. The actual contribution of the root was calculated assuming that the water content in the soil in front of and behind the root is equal to that of the soil at the sides of the roots in the radiographs. The thickness of water in the root, $L_{H_2O}^{root}$, can be calculated as:

$$L_{H_2O}^{root} = \frac{(\mu_{H_2O}d_{H_2O})_{root} - (d_{tot} - d_{root}) \times (\mu_{H_2O}d_{H_2O})_{soil}}{\mu_{H_2O}} \quad (4)$$

where $(\mu_{H_2O}d_{H_2O})_{root}$ is the attenuated neutron beam in the centermost pixel of the root, $(\mu_{H_2O}d_{H_2O})_{soil}$ is the attenuated neutron beam in the soil near the root, d_{root} is the root diameter. We used the rehydration rate of roots to calculate the flow rate of water into the roots. We calculated the volumetric flow of water into the roots per unit of root length, J_r^{root} [$\text{m}^3 \text{ s}^{-1} \text{ m}^{-1}$ or $\text{m}^2 \text{ s}^{-1}$], from the changes in the

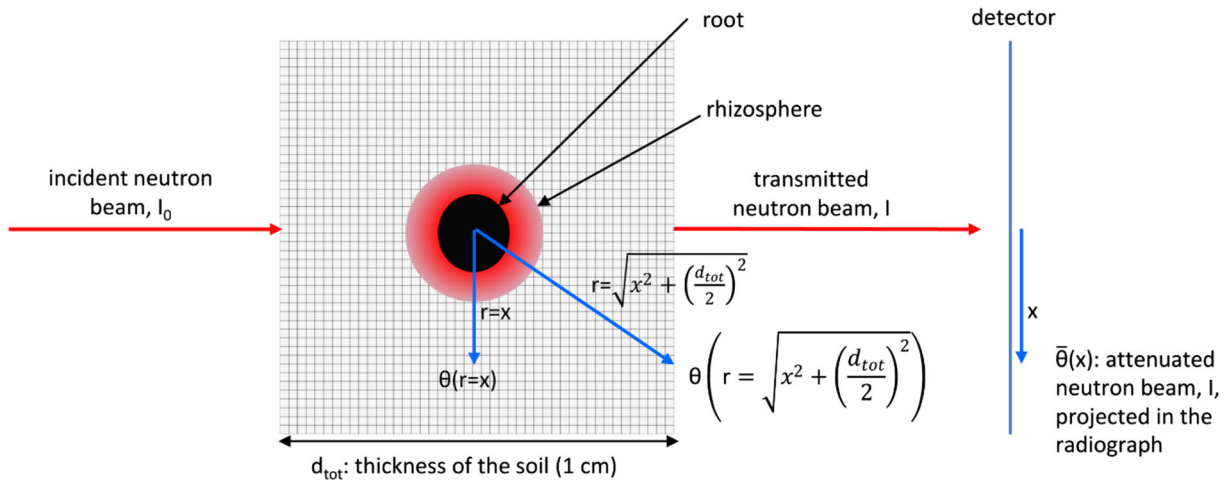


Fig. 1 Schematic cross-section of root (black), rhizosphere (red) and bulk soil (light gray). Neutron radiography gives the water content averaged along the sample thickness. In the pixels next to the root, the imaged water content is the average water content $\bar{\theta}$ in

bulk soil and rhizosphere. The actual profile of water content as a function of distance from the root surface $\theta(r)$ can be estimated from $\bar{\theta}(x)$ according to Eq. (3)

water-filled root volume per unit of root length, V [$\text{m}^3 \text{m}^{-1}$], according to:

$$J(t) = \frac{\partial V(t)}{\partial t} \quad (5)$$

where

$$V(t) = \pi \left(\frac{L_{H_2O}^{root}(t)}{2} \right)^2 \quad (6)$$

where V is the water-filled root volume per root length [$\text{m}^3 \text{m}^{-1}$]. Note that Eqs. (5) and (6) include the effects of both, swelling of the root tissue and refilling of the root tissue. To calculate $V(t)$ we assumed that the water in the root tissue was contained in a cylinder oriented perpendicularly to the direction of the neutron beam and that the axial transport of water along the root was negligible. The water content in root and soil were averaged along the selected root segments, which were approximately 4–6 cm long.

Root pressure measurement

In a parallel experiment, we used a root pressure probe (Frensch and Steudle 1989) to monitor the xylem pressure in the stem of plants during irrigation. The root pressure probe consists of a pressure transducer connected to the excised stem of the plants via a glass capillary. A tight connection between the cut stem and the pressure probe

was achieved using a silicone material (Xantopren L blue, Heraeus Kulzer GmbH, Germany) that was carefully pressed around the stem. Half of the capillary (diameter of 600 μm) was filled with silicon oil and the other half of the capillary facing the root was filled with water. The meniscus forming between water and oil allowed monitoring the direction of the water flow. The root pressure probe measures the hydrostatic pressure in the xylem of the stem.

We grew three lupins with the same conditions used in the neutron radiography measurements. We stopped irrigating the plants 6 days before the measurements. We started the measurement when the plants were 30 days old and the soil was at a water content of 0.02–0.04 $\text{m}^3 \text{m}^{-3}$. The measurements started at nighttime when plants were not transpiring and the water potential in the root was expected to approach that of the soil.

The shoots were excised using a wet razor blade and the cut ends of the roots were quickly inserted and sealed into the root pressure probe, which was initially filled with water. During the connection, we carefully controlled that no air bubbles were present in the measuring device. When the pressure measured by the pressure probe was constant (it took around 1 h), we irrigated the plants from the top with a constant flow rate of 1 ml min^{-1} . We monitored the root pressure for a period of 3 h after irrigation. The experiments were repeated for three plants.

The root pressure probe measures only the hydrostatic component of the water potential in the xylem, $\Psi_{p,x}$ [MPa]. The total water potential $\Psi_{w,x}$ [MPa] depends also on the osmotic potential $\Psi_{s,x}$ [MPa] induced by the

presence of the solutes in the xylem, and on the gravitational potential. We neglected the latter, as it was constant among the samples and small compared to the other components (in particular when the soil was dry). It follows that the total water potential was:

$$\psi_{w,x} = \psi_{p,x} + \psi_{s,x} \quad (7)$$

where the subscript x refers to the xylem.

To calculate the water potential in the xylem we made the following assumptions: 1) long after irrigation, root and soil reached equilibrium and had an equal water potential of $\psi_{w,x}^{fin}$; 2) The osmotic potential in the xylem $\psi_{s,x}$ did not change during the experiments. It follows that:

$$\psi_{s,x} = \psi_{w,x}^{fin} - \psi_{p,x}(t \rightarrow \infty) \quad (8)$$

where $\psi_{p,x}(t \rightarrow \infty)$ is the value of the hydrostatic component of the water potential long after irrigation. The root water potential in the xylem during the irrigation was calculated by summing up the value of $\psi_{s,x}$ obtained from Eq. (8) with the measured hydrostatic pressure. Note that the assumption of constant osmotic potential during the rewetting experiment should deserve additional attention and further studies.

Overall hydraulic conductivity of the rhizosphere-root continuum

The flow of water from the outermost part of the rhizosphere into the root xylem has to overcome the hydraulic resistances of the rhizosphere and the root. The total hydraulic conductance of the pathway across rhizosphere and root tissue till the root xylem can be described as:

$$J = K_{tot}(\psi_{w,soil} - \psi_{w,x}) \quad (9)$$

where J is the volumetric flow rate of water into the root per unit of root length [$\text{m}^2 \text{s}^{-1}$], K_{tot} is the total hydraulic conductance of the water pathway from the rhizosphere into the root xylem per root length [$\text{m}^2 \text{s}^{-1} \text{MPa}^{-1}$], $\psi_{w,soil}$ is the water potential in the bulk soil [MPa], $\psi_{w,x}$ is the water potential in the root xylem [MPa].

We used the swelling rates of the roots obtained from neutron radiographs to calculate the flux of water into the roots, assuming that the change in the water volume in the roots was equal to the net flow of water into the roots, and that the axial flow along the roots was small (Eq. 5). The soil water potential during the infiltration experiment

($\psi_{w,soil}$) was estimated based on the soil water content obtained from neutron radiographs and the measured soil water retention curve of the bulk soil. The soil retention curve of a disturbed sample was determined in the lab using a pressure plate apparatus method. The root water potential ($\psi_{w,x}$) was derived from the root pressure probe experiment as discussed above. Using all this information, we calculated K_{tot} using Eq. (9).

Hydraulic conductivity of the rhizosphere

The total hydraulic conductance of the pathway of water across the rhizosphere and the root can be written as:

$$\frac{1}{K_{tot}} = \frac{1}{K_{rhizo}^{eff}} + \frac{1}{K_{root}} \quad (10)$$

where K_{rhizo}^{eff} is the effective hydraulic conductance of the rhizosphere [$\text{m}^2 \text{s}^{-1} \text{MPa}^{-1}$] and K_{root} is the hydraulic conductance of the root [$\text{m}^2 \text{s}^{-1} \text{MPa}^{-1}$]. The total conductance K_{tot} was calculated using the harmonic mean. The harmonic mean gives the conductance of the flow through a series of elements. In our case, the elements in series are the rhizosphere and the root.” Note that here all the conductances refer to a unit of root length. In this experimental set-up we could not distinguish between the effect of roots and rhizosphere and, consequently, we could not calculate K_{root} and K_{rhizo}^{eff} . However, to have an approximated estimation of K_{rhizo}^{eff} we made the following assumptions.

We took the value of K_{root} for a lupin root based on the data of Zarebanadkouki et al. (2013), who reported the radial flux of water into different parts of lateral roots of lupin. According to Eq. (9), the total hydraulic conductance of a root can be calculated from the total flow of water in each root and the gradient in water potential between the soil and the root xylem. We used the measured radial fluxes of water into 12–14 and 8–10 cm long lateral roots located in the upper root zone system (depth of 2–9 cm). Additionally, we used a pressure chamber to measure the water potential in the xylem of lupin plant that had the same age, growing conditions and transpiration rate. We carefully removed the root from soil and sealed it in the pressure chamber with the cut stem protruding through the lid of the chamber. Then, by slowly pressurizing the system, we measured the pressure when water started to flow out from the xylem. We referred to this pressure as minus the xylem water potential. The xylem water potential at midday was -0.25 MPa . This value was

taken as the water potential in the root xylem; note that here we assumed that the pressure dissipation along 9 cm of the taproot was negligible. Using the soil water potential obtained from the water retention curve and the radial flow of water reported by Zarebanadkouki et al. (2013), we calculated K_{root} $1.68 \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ MPa}^{-1}$ (average of lateral roots with a length of 12–14 and 8–10 cm). Here the calculated value of K_{root} is divided by the root length for comparison with the conductances defined in Eqs. (9 and 10).

An alternative way to estimate K_{root} is explained in the following lines: let us assume that towards the end of the experiment, when the rhizosphere was rewetted, the flow of water into the roots was mainly controlled by the hydraulic conductance of the roots. From this assumption, it follows that:

$$K_{tot}(t = n) = K_{root} \quad (11)$$

where $t=n$ refers to a time when the rhizosphere is rewetted. We expect that the value of $K_{root} = 1.68 \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ MPa}^{-1}$ obtained from the data of Zarebanadkouki et al. (2013) is close to the value of K_{root} obtained from Eq. (11).

Rearranging Eq. (10), K_{rhizo}^{eff} is given by:

$$\frac{1}{K_{rhizo}^{eff}} = \frac{1}{K_{tot}} - \frac{1}{K_{root}} \quad (12)$$

Under steady state conditions, the flow of water across the rhizosphere can be described as:

$$J = K_{rhizo}^{eff} (\psi_{w,soil} - \psi_{w,rs}) \quad (13)$$

where $\psi_{w,rs}$ is the water potential in at the root surface [MPa]. Assuming a cylindrical symmetry, the flow of water across the rhizosphere towards the root can be described as (Nobel and Cui 1992a):

$$J = 2\pi k_{rhizo} \frac{(\psi_{w,soil} - \psi_{w,rs})}{\ln\left(\frac{r_{rhizo}}{r_{root}}\right)} \quad (14)$$

where r_{root} is the root radius, r_{rhizo} is the radius of the outermost part of rhizosphere and k_{rhizo} is the hydraulic conductivity of the rhizosphere [$\text{m}^2 \text{ s}^{-1} \text{ MPa}^{-1}$]. According to Carminati et al. (2010) and Moradi et al. (2011), we took $r_{rhizo} = 1.85 \text{ mm}$ (note that this value is the sum of the root radius and the rhizosphere thickness). If we express the water potential in pressure head [m], k_{rhizo} has the units of m s^{-1} . Based on Eqs. (13 and 14), the

hydraulic conductivity of the rhizosphere can be written as:

$$k_{rhizo} = \frac{1}{2\pi} K_{rhizo}^{eff} \ln\left(\frac{r_{rhizo}}{r_{root}}\right) \quad (15)$$

Note that in these equations we assumed a steady-state scenario and a uniform rhizosphere conductivity through the rhizosphere. Despite the several assumptions and approximations (in particular the assumption of a constant K_{root}), Eqs. (12–15) gives a first estimation of the rhizosphere conductivity during the drying/rewetting process.

Results

We monitored the water content in soil and roots during irrigation of five samples using a time-series neutron radiography. To illustrate the experiment and the approaches taken to quantify the results, we first present the results of one sample in details. Then, we present the hydraulic conductivities of 12 roots from a total of five plants. Figure 2 shows the radiographs of one sample at different times after irrigation. The radiographs are a close-up of the original field of view of $15.5 \times 15.5 \text{ cm}$. The sample was irrigated from the top with a constant infiltration rate of 1 ml min^{-1} . Figure 2a and c show the radiographs at 30 and 120 min after irrigation, respectively. In these images, darker gray values correspond to the higher water contents. After irrigation, the water content in the bulk soil increased, while a tiny region around the roots remained remarkably drier (Fig. 2b). This indicates that the bulk soil was conductive enough to be rewetted quickly while the rewetting of the rhizosphere was restricted. Figure 2d shows the ratio between the actual radiographs at 120 and 30 min after irrigation. We chose the radiograph taken 30 min after irrigation because at this time the wetting front covered all the roots selected for image analysis. Figure 2d shows the changes in water content from 30 to 120 min after irrigation. In this image, the darker gray values correspond to a bigger increase in water content. This image shows that the largest changes in water content during the time interval between 30 and 120 min occurred in the roots and their rhizosphere.

After root segmentation, we quantified the water content in soil, rhizosphere and roots according to Eqs. (1–3). Figure 3a shows the profile of water content as a function of distance from the surface of a selected root from a

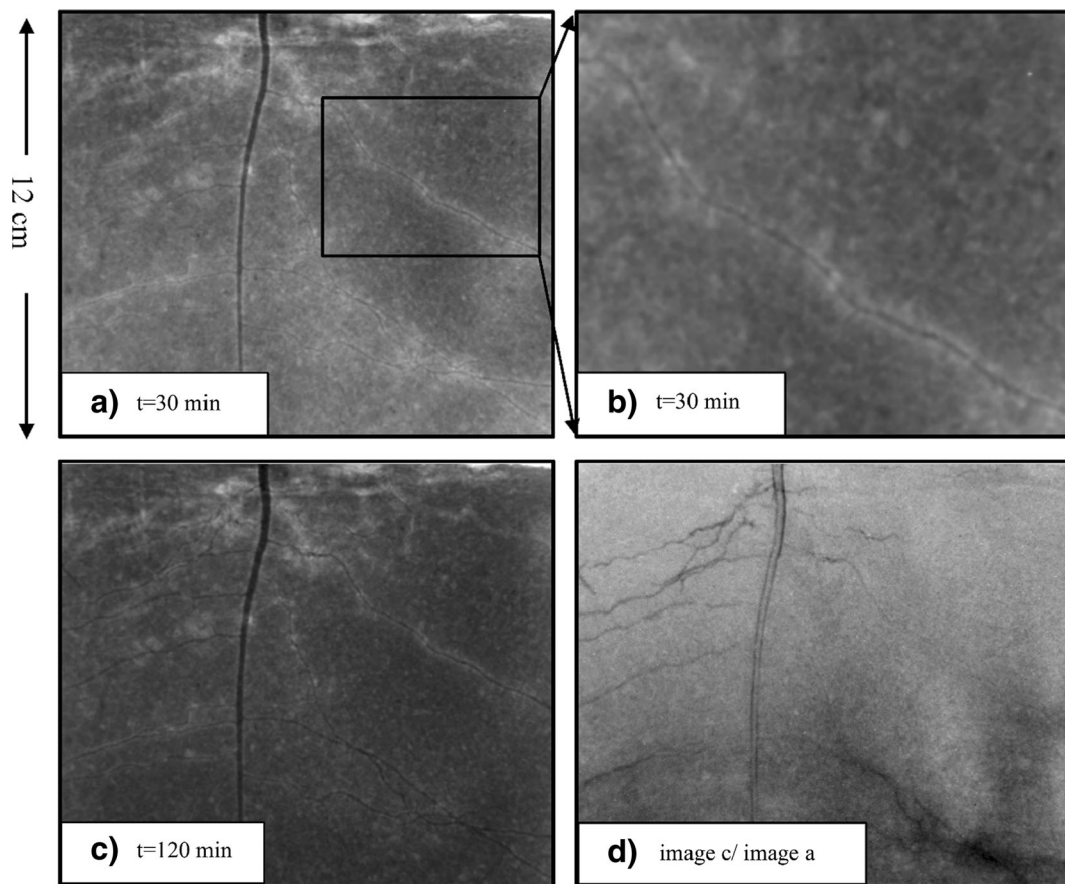


Fig. 2 **a–c)** Neutron radiographs of roots and soil water content distribution in a sample irrigated from the top at different times after irrigation. The darker is the gray value the wetter is the soil. **b)** Close up of Fig. 2a. The bright region around roots indicates a slow

rewetting of the rhizosphere. **d)** Ratio between the radiographs taken 120 and 30 min after irrigation. The darker is the image the bigger is the increase in water content after irrigation. Before irrigation, the soil water content was $0.02\text{--}0.04\text{ m}^3\text{ m}^{-3}$

sample irrigated with a constant infiltration rate of 1 ml min^{-1} from the top (constant irrigation rate experiment). The profiles of water content were averaged along root segments of 4–6 cm long. The results showed that before irrigation soil water content in the soil close to the root (the rhizosphere) was higher than in soil far from the root (the bulk soil). Immediately after irrigation the soil water content in the bulk soil increased while the soil near the root remained temporarily dry and it rewetted at a much slower rate.

Two plants were irrigated with an infiltration rate of 10 ml min^{-1} for 6 min (pulse irrigation rate experiment). In this set-up, we injected water into the soil from the top of the sample for a period of 6 min and then we stopped irrigation. Figure 3c shows the profile of water content as a function of distance from the surface of a selected root of a sample irrigated with pulse irrigation. As in Fig. 3a, the soil near the root remained initially dry and it

rewetted more slowly than the bulk soil. It is interesting to note that when irrigation finished and the water content in the soil far from the root started to decrease due to water redistribution to deeper layers, the water content in the soil near the root continued to increase. Note that only the upper part of the sample was rewetted during the 6 min of irrigation and after irrigation water redistributed throughout the sample and the water content in the upper part decreased.

Figure 3b shows the average water contents in the rhizosphere and the bulk soil of different roots of a sample irrigated with constant irrigation. The rhizosphere water content was calculated as the average water content in the first 1.5 mm near the root surface. The value of 1.5 mm was chosen as an indicative value for the rhizosphere thickness (Moradi et al. 2011). However, it should be kept in mind that the rhizosphere properties change continuously from the bulk soil to the root surface and that there is no

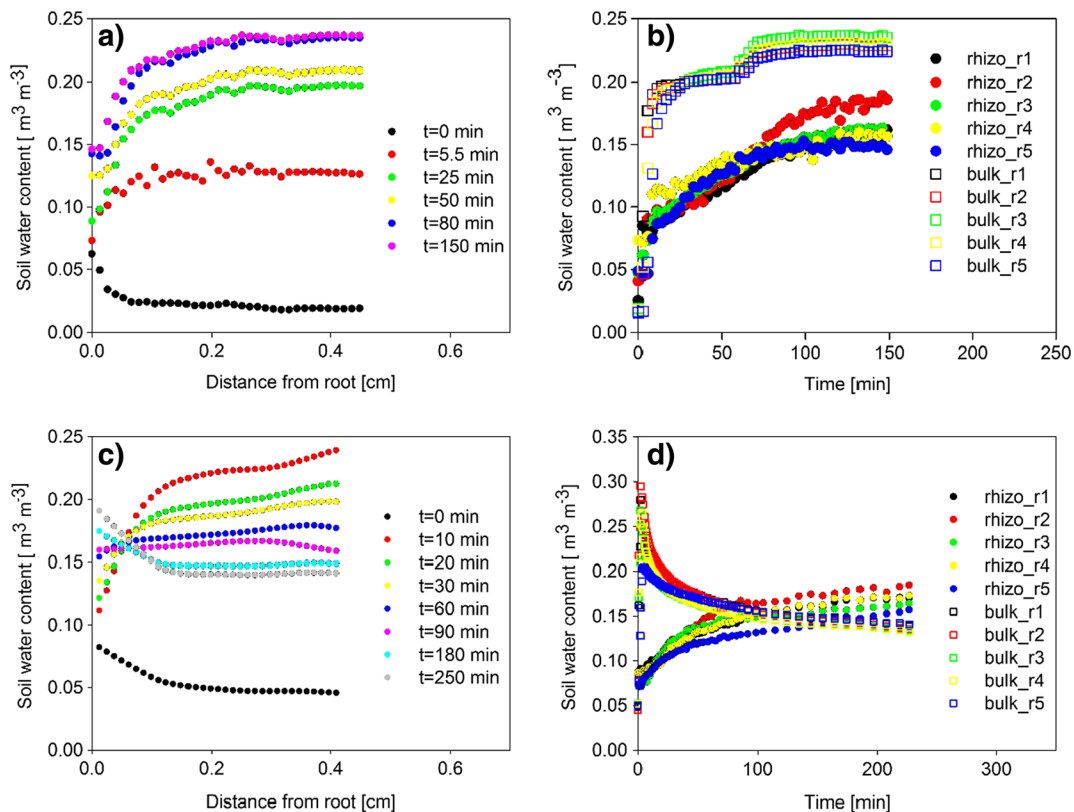


Fig. 3 **a)** Profiles of water content as a function of distance to one root of the plant shown in Fig. 2. **b)** Average water content in the rhizosphere of five roots of the same plant shown in Fig. 2 compared to the water content in the bulk soil. **c)** Profiles of water content as a function of distance to one root of one plant during the

pulse irrigation rate experiment. **d)** Average water content in the rhizosphere of five roots of the plant shown in the subplot **c)** compared to the water content in the bulk soil. Note that all the profiles are averaged along root segments with length of 4 to 6 cm located at a distance of 5 to 10 cm from the root tip

sharp boundary between bulk soil and rhizosphere as seen in Fig. 3a. Calculation of the average rhizosphere water content was intended to illustrate the differences between the bulk soil and the soil near the root. Figure 3b shows that the rhizosphere rewetted more slowly than the bulk soil. A similar behavior was observed in a sample irrigated with pulse irrigation (Fig. 3d). As mentioned above, the water content in the rhizosphere continued to increase even while the bulk water content was decreasing.

Figure 4a shows the changes in the water-filled volume of five roots of the plant presented in Fig. 2. The water-filled root volume was calculated according to Eq. (6) and it refers to the water-filled root volume per root length. Note that the changes in water-filled root volume depend on the swelling of the root tissue and the refilling of the air-filled space in the root tissue. Root rehydration was slow during the first 10–15 min after infiltration. Afterwards, the rehydration rate increased until it reached a constant value. After approximately

100 min, the water-filled root volume reached a constant value, indicating that root and soil reached hydraulic equilibrium. We observed a final root swelling of $63 \pm 18\%$ ($n=5$). In Fig. 4b we showed the root volumes divided by the final root volumes and the best fit of the measurements (red line).

Figure 4c shows the change in water potential in the xylem during irrigation. Although the curves showed similar temporal dynamics, the initial value varied significantly among the samples. This was probably caused by the different initial conditions of the plants before irrigation. In fact, we irrigated the samples when the soil was at a very low water content ($0.02\text{--}0.04\text{ m}^3\text{ m}^{-3}$), when small changes in water content correspond to very large differences in soil water potential (see the water retention curve in the supplementary information). The red line indicates the best fit among the three measurements.

In Fig. 4d we plotted the relation between root volume and root water potential based on the best

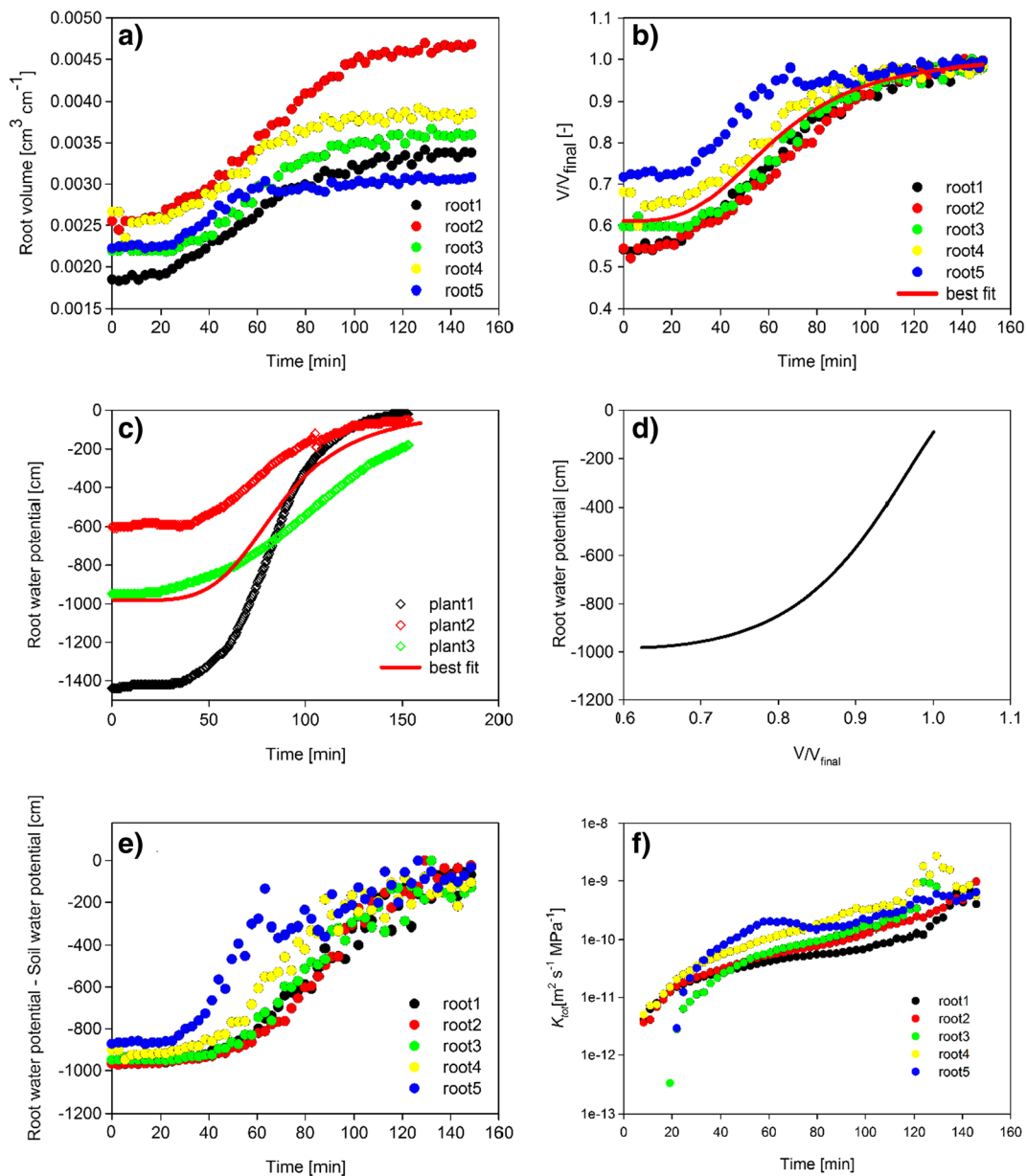


Fig. 4 **a)** Volume of five different roots after irrigation obtained from neutron radiography. The volumes are volumes per root length. The roots in Fig. 4a, b, d, e, and f belong to the sample shown in Fig. 2. **b)** Normalized root volume after irrigation and best fit (red line). The root volumes are normalized to the final root volume. **c)** Xylem water potential in three different plants measured with the root pressure probe and the best fit (red line). **d)** Relation between xylem water potential and root volume. The relation is

derived from the best fits of the normalized root volume and of the xylem water potential. **e)** Gradients in water potential between roots and the adjacent bulk soil during irrigation. Root water potentials were derived from Fig. 4a and d. The bulk soil water potential was estimated from the soil water retention curve and the measured bulk soil water content. **f)** Total hydraulic conductance of the rhizosphere-root continuum as a function of time after irrigation

fit of the measurements. To derive this relation, we plotted the best fit of the normalized root volume versus the best fit of the xylem water potential. Figure 4d shows an exponential relation between

root volume and water potential. We used this relation to estimate the root water potential of different plants from the data of their root volumes during irrigation.

We calculated the total hydraulic conductance of the rhizosphere-root continuum K_{tot} according to Eq. (9). We used the root swelling rate (Fig. 4a) to calculate the flow of water into roots. The gradients in water potential between soil and roots were calculated using the soil water retention curve (supplementary information) and the relation between root water potential and root volume (Fig. 4e). Figure 4f shows the changes in K_{tot} after irrigation for five roots of the plant shown in Fig. 2. Shortly after irrigation the K_{tot} was low and it increased of two orders of magnitude over time.

To estimate K_{tot} we used the pressure volume curve obtained from the best fit of the measurements of root volumes (Fig. 4a) and root pressures (Fig. 4c). The measurements of root pressure during irrigation showed some variabilities. To assess the effect of these variations on the estimation of K_{tot} , we repeated the calculations using the pressure curves obtained from different plants. In Fig. 5a we plotted the best fit of the root volumes (Fig. 4b) versus the root pressures measured for the three plants. Then we calculated K_{tot} based on the three newly calculated pressure volume curves. The values of K_{tot} slightly varied (Fig. 5b), but the trends and relative changes in K_{tot} were rather similar.

In Fig. 6a we plotted K_{tot} for 12 roots of five different plants as a function of the water content in the rhizosphere. Immediately after irrigation $K_{tot}=1.36\pm1.09\times10^{-11}\text{ m}^2\text{ s}^{-1}\text{ MPa}^{-1}$ (average of the first 10 min after irrigation). Afterwards, K_{tot} increased and reached a value of $5.02\pm2.13\times10^{-10}\text{ m}^2\text{ s}^{-1}\text{ MPa}^{-1}$ at 2.5 h after irrigation. In Fig. 6a we also plotted K_{root} obtained from Zarebanadkouki et al. (2013), who measured the radial flux of water into lateral roots of lupines and the xylem water potential, obtaining a value of $K_{root}=1.68\times10^{-9}\text{ m}^2\text{ s}^{-1}\text{ MPa}^{-1}$. Shortly after irrigation K_{tot} was two orders of magnitude smaller than K_{root} . With time K_{tot} increased and approached K_{root} .

The hydraulic conductivity of the rhizosphere k_{rhizo} , estimated using Eqs. (12–15), is plotted in Fig. 6b. During the first 10 min after irrigation $k_{rhizo}=3.87\pm1.91\times10^{-14}\text{ m s}^{-1}$ and it increased to $k_{rhizo}=1.09\pm1.64\times10^{-12}\text{ m s}^{-1}$ at 2.5 h after irrigation.

Discussion

Consistently with previous observations (Carminati et al. 2010; Moradi et al. 2011, 2012), we found that the rhizosphere held more water than the bulk soil

during drying of soil by plants and it stayed temporarily drier than the bulk soil after irrigation. Our data also showed that as the rhizosphere rewetted, the roots slowly rehydrated. By means of image processing, we quantified the rewetting rate of the roots and we used this information to calculate the total hydraulic conductance of the rhizosphere-root continuum K_{tot} .

To understand the impact of this result on root water uptake, it is interesting to compare K_{tot} to typical values of root hydraulic conductance. K_{root} of a lupin root was calculated based on the data of Zarebanadkouki et al. (2013), who measured the radial flux of water into lateral roots of lupins and the xylem water potential, obtaining $K_{root}=1.68\times10^{-9}\text{ m}^2\text{ s}^{-1}\text{ MPa}^{-1}$ (conductance per root length). We found that as the rhizosphere turned hydrophobic K_{tot} was two orders of magnitude smaller than the hydraulic conductance of lupin roots in wet soils. We further calculated the hydraulic conductivity of the rhizosphere, k_{rhizo} . We found that the hydraulic conductivity of the rhizosphere was $k_{rhizo}=3.87\pm1.91\times10^{-14}\text{ m s}^{-1}$ shortly after irrigation and it increased of two orders of magnitude in 2 to 3 h after irrigation. For this estimation, it is important to remember that we assumed the K_{root} did not change during the drying/wetting cycles; an assumption that will be discussed later on. First, let us focus on the total conductance of the root-rhizosphere continuum.

The main results of this study is that K_{tot} was two orders of magnitude smaller than the hydraulic conductance of lupin roots in wet soils and it took 2 to 3 h for K_{tot} to recover. One explanation of the low hydraulic conductance of the root-rhizosphere continuum after drying and rewetting is the temporary water repellency of the rhizosphere. Moradi et al. (2012) showed that the rhizosphere of lupin roots turned hydrophobic after severe drying. Similarly, Hallett et al. (2003) measured a decreased water sorptivity in the rhizosphere of barley. Read and Gregory (1997) showed that presence of lipids in the mucilage of maize, lupin and wheat decreased the surface tension of water. These lipids, although present in a low concentration, may explain the hydrophobicity of mucilage (Moradi et al. 2012). The theory of water flow through soils mixed with mucilage is described in Carminati (2012) and Kroener et al. (2014).

A second explanation of the low hydraulic conductance of the root-rhizosphere continuum is the presence of air-filled gaps at the root-soil interface (Nobel and Cui 1992b; Nye 1994; Carminati et al. 2009). Carminati et al. (2012) found that the gaps formed around the roots

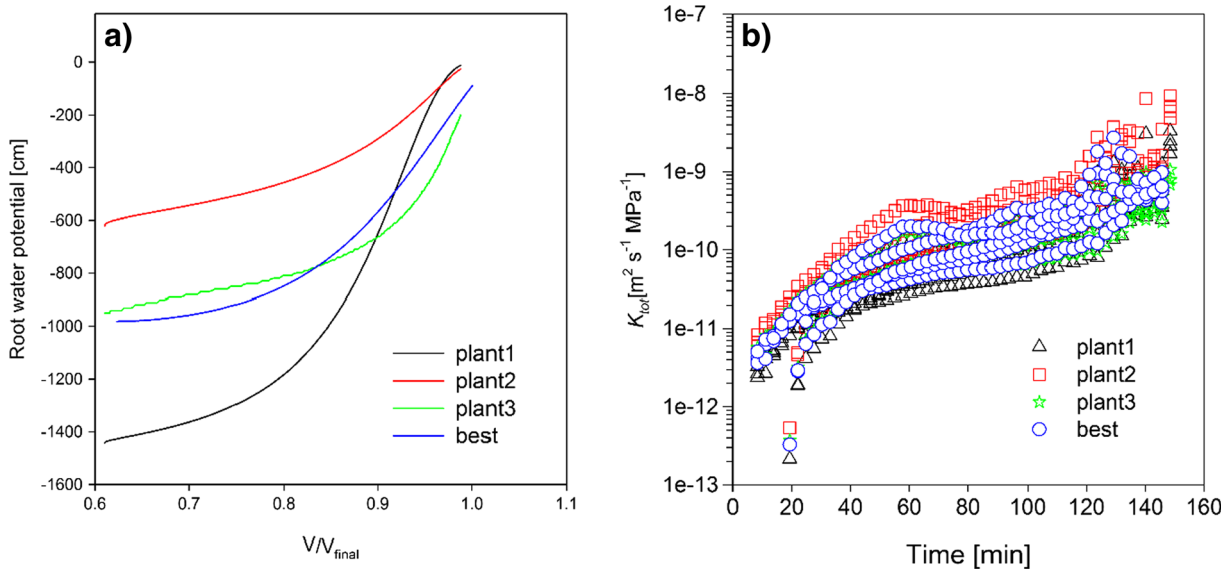


Fig. 5 **a)** Pressure volume curves obtained plotting the best of fit of root volumes (Fig. 4b) as a function of the root pressures measured in three different plants (plus the best fit). **b)** Total hydraulic

conductance of the rhizosphere-root continuum (K_{tot}) as a function of time after irrigation calculated using the different pressure volume curves shown Fig. 5a

of lupins at soil matric potentials between -10 and -50 kPa. Gaps around the tap root were large and reached values of 0.1 – 0.35 mm. However, the gaps around the lateral roots were smaller with an average size of 0.025 mm in the upper soil region. Based on the Young-Laplace equation, such gaps should be water-filled for soil matric potentials higher (less negative) than -12 kPa (assuming a pore radius of 0.0125 mm

for the gaps). During our irrigation experiments, the soil matric potential was about -6 kPa. Neglecting the effect of mucilage in increasing the contact angle, we would expect that the gaps around the lateral roots were water-filled. Consequently, the loss of contact between lateral roots and the soil, by itself, should have not limited root rehydration upon irrigation. However, when coupled with the rhizosphere water repellency, gaps could have

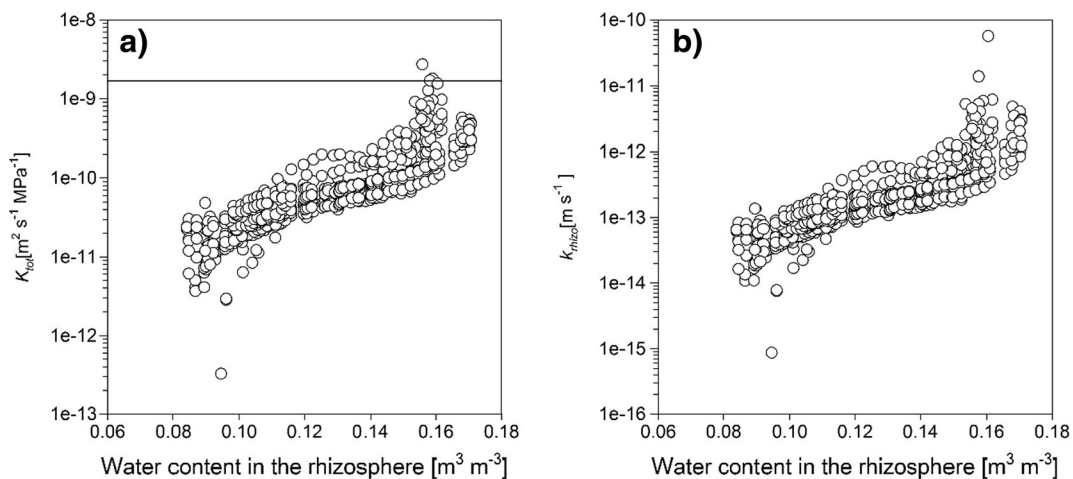


Fig. 6 **a)** Total hydraulic conductance of the rhizosphere-root continuum K_{tot} as a function of water content in the rhizosphere (dots) and hydraulic conductance of a lupine root estimated based on the fluxes reported in Zarebanadkouki et al. 2013 (line). **b)** The

hydraulic conductivity of the rhizosphere (k_{rho}) as a function of water content in the rhizosphere as obtained from Eqs. (12–15). The data are obtained from 12 roots of five different plants

further limited the water flow across the root-soil interface.

Our approach to calculate the total hydraulic conductance of the rhizosphere-root continuum is quite straightforward. On the contrary, the calculation of hydraulic conductivity of the rhizosphere required further assumptions. We assumed that the hydraulic conductance of lupin roots, K_{roots} , did not change during the drying/rewetting cycles and we estimated K_{root} from experiments with roots grown in a wet soil. Doing so, we implicitly assumed that the major reduction in conductivity occurred in the rhizosphere. These assumptions cannot be confirmed based on our actual data. Additional complementary measurements of K_{root} during drying and rewetting would be needed to separate the effects of root and rhizosphere on the total hydraulic conductance.

However, it should be mentioned that the large recovery of K_{root} after rewetting is not easily explainable with a change in root resistance. Roots change their permeability in response to water stress by further formation of casparian bands, deposition of suberin lamellae at the endodermis and regulation of aquaporin activity (Steudle and Peterson 1998). Among these factors, only aquaporin activity is reversible and can explain the reduction and recovery of K_{tot} . However, even after complete blockage of aquaporin, root permeability did not decrease of more than 80 % (Martre et al. 2001; Bramley et al. 2009; Knipfer et al. 2011). Cavitation in the xylem could also explain the reduction in hydraulic conductivity. However, the hydraulic conductivity was calculated based on the swelling rate of the roots, which depended only on the radial flux of water across the root-soil interface.

What are the implications of this study? A likely consequence of the low hydraulic conductance of the root-rhizosphere interface after drying is a temporary reduction in root water uptake after irrigation. Indeed, Zarebanadkouki and Carminati (2014) observed that the flow of water into roots was 4 to 8 times reduced as the rhizosphere turned hydrophobic. The reduction was more significant in the mature root segments, where the rewetting rate of the rhizosphere is lower (Carminati and Vetterlein 2013). Zarebanadkouki and Carminati (2014) measured the average flux of water into the lateral roots during 90 min after irrigation. Similarly, our data showed that for an average of 90 min after irrigation, the hydraulic conductivity of the rhizosphere was 7 ± 4 ($n=12$) times smaller than the calculated hydraulic conductivity of the root.

At the first glance, the temporary low hydraulic conductance of the root-soil interface seems to be harmful for plants, as it may limit the flow of water into the roots that experienced a prolonged water stress. However, this phenomenon has broader and more complex consequences: a hydrophobic rhizosphere with a low hydraulic conductivity may build up an additional resistance to prevent the back flow of water from the roots into the soil. For a deep-rooted plant, Zarebanadkouki et al. (2013, 2014) showed that the basal segments of roots near the soil surface are initially more involved in root water uptake. Similarly, Ahmed et al. (2015) showed the same trend for a 2-weeks-old maize (*Zea mays* L.). Therefore, the upper soil layers dry quickly, while deeper roots have access to water from the wet subsoil. Hydraulic isolation of the upper roots will make the plant feeling only the wet soil layers, will shift the uptake to deeper roots and will limit the back flow of water from the upper roots into the dry soil.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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